Image Classification.

3ta turi mavjud:

Binary → 1ta label → Cat / Dog
Multiclass → 1ta label → more than 2 classes
Multilabel → many labels → Person + Car.

Binary:

→ Faqat 2ta output ya'ni 2tadan bittasini tanlaymiz.
=> Healthcare → disase vs no disease.
=> manufacturing - defective vs good.
=> Security - mask vs no mask.
=> Finance - fraud vs not fraud.

Multi class classification (one label only)

→ output kop faqat bittasi tanlanadi.
=> Autonomous driving - traffic sign recognition.
=> Agriculture - plant disase type.
=> Retail - product category.
=> Education - Handwritten digit recognition.

Multi label (Multi output).

→ Output kop, javob ham kop.
Har bir image koplab labellarga ega

=> Social media - Image tagging (#person, #car, #tree)
=> Healthcare - Multiple disase in one scan.
=> E-commerce - Product attributes (color, type, style)

Image classification system architecture.

(1) Data preparation.

=> Data Acquisition - Gathering and labeling raw images.

=> Preprocessing - Resizing, cropping, normalization of pixels.

=> Augmentation - Creating variations to improve generalization.

(2) Architecture

=> Backbone - Extracting features from images.

=> Neck/Head - Aggregating features and producing output.

=> Softmax - Converting outputs into probabilities.

(3) Training

=> Optimization - updating model weights based on loss.

=> Validation - Monitoring performance during training.

=> Evaluation - Testing on unseen data.

(4) Deployment

=> Optimization - Making model faster and lighter.

=> Inference - Running predictions on new images.